EXPERIMENT NO:-04

Aim: Apply DML Commands for the specified system.

Theory:

The structured query language (SQL) commands deal with the manipulation of data present in the database that belongs to the DML or Data Manipulation Language. This includes most of the SQL statements.

Examples of DML

The examples of DML in the Database Management System (DBMS) are as follows –

- SELECT – Retrieve data from the database.
- INSERT – Insert data into a table.
- UPDATE – Update existing data within a table.
- DELETE – Delete records from a database table.

Syntax for DML Commands

The syntax for the DML commands are as follows –

- **INSERT**

Insert command is used to insert data into a table.

The syntax for insert command is as follows –

Syntax

Insert into <table_name> (column list) values (column values);

For example, if we want to insert multiple rows to the Employee table, we can use the following command –

Example

Insert into Employee(Emp_id, Emp_name) values (001, “ bhanu”);

Insert into Employee(Emp_id, Emp_name) values (002, “ hari”);

Insert into Employee(Emp_id, Emp_name) values (003, “ bob”);

- **SELECT**

Select command is used to retrieve data from the database.

The syntax for the select command is as follows –

Syntax

SELECT * from <table_name>;

For example, if we want to select all rows from the Employee database, we can use the following command –

Example

SELECT * from Employee;

- **DELETE**

Delete command is used to delete records from a database table.

The syntax for the delete command is as follows -

Syntax

Delete from <table_name>WHERE condition;

For example, if we want to delete an entire row of employee id 002, we can use the following command –

Example

DELETE from Employee WHERE Emp_id=002;

- **UPDATE**

Update command is used to update existing data within a table.

The syntax for the update command is as follows -

Syntax

UPDATE <table_name> SET column_number =value_number WHERE condition;

For example, if we want to update the name of the employee having the employee id 001, we can use the command given below –

Example

UPDATE Employee SET Emp_name= Ram WHERE Emp_id= 001;

Application of DML commands on Specified Case study with Input & Output:

Q1 : Display all pdf details ?
Ans : select * from PDF;

```
SELECT * FROM PDF;
```

Result Grid

pdf_id	title	description	file_path	upload_date	admin_id	category_id
1	DBMS Basics	Intro notes	/files/dbms.pdf	2024-01-10	1	1
2	CN Short Notes	Networking	/files/cn.pdf	2024-01-12	2	2
3	OS Concepts	Processes	/files/os.pdf	2024-01-15	3	3
4	Java Guide	OOP	/files/java.pdf	2024-01-18	4	4
5	Python Handbook	Basics	/files/python.pdf	2024-01-20	5	5
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Q2 : Display all title from the pdf ?
Ans : select title from PDF;

```
SELECT title FROM PDF;
```

Result Grid

title
DBMS Basics
CN Short Notes
OS Concepts
Java Guide
Python Handbook

Q3 : Display all first name from the admin ?
Ans : select first_name from Admin;

```
1 •  SELECT first_name FROM Admin;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name
Ramesh
Suresh
Anita
Karan
Meena

Q4 : Display all title from the pdf where id=1 ?
Ans : select title from PDF where category_id=1;

Don't Limit

```
1  SELECT title FROM PDF WHERE category_id = 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

title
DBMS Basics

Q5 : Display all access log where action=view ?
Ans : select * from Acess_log where action='View';

```
SELECT * FROM Access_Log WHERE action = 'VIEW';
```

login_id	access_date	action	pdf_id	user_id
1	2024-02-01 10:00:00	VIEW	1	1
3	2024-02-02 11:00:00	VIEW	3	3
5	2024-02-04 01:00:00	VIEW	5	5
NULL	NULL	NULL	NULL	NULL

Q6 : Display all first name from user where user id=3 ?
Ans : select first_name from User where user_id=3;

```
SELECT first_name FROM User WHERE user_id = 3;
```

first_name
Rohit

Q7 : Display all title from the pdf where id is in between 2 and 4 ?
Ans : select title from PDF where pdf_id between 2 and 4;

```
SELECT title FROM PDF WHERE pdf_id BETWEEN 2 AND 4;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

title
CN Short Notes
OS Concepts
Java Guide

Q8 : Display all user details ?
Ans : select * from User;

```
SELECT * FROM User;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

user_id	first_name	middle_name	last_name	mobile_no	password
1	Amit	NULL	Verma	8000000001	newpass
2	Neha	NULL	Singh	8000000002	user2
3	Rohit	NULL	Kumar	8000000003	user3
4	Priya	NULL	Shah	8000000004	user4
5	Rahul	NULL	Mehta	8000000005	user5
NULL	NULL	NULL	NULL	NULL	NULL

User 11

Q9 : change category name to AI for id 5?
Ans : Update Category
Set category_name='AI'
Where category_id= 5;

```
UPDATE Category
SET category_name = 'AI'
WHERE category_id = 5;

SELECT * FROM Category;
```

category_id	category_name
5	AI
2	Computer Networks
1	DBMS
4	Java
3	Operating System
NULL	NULL

Q10 : change mobile number to 999999999 for id 3?

Ans : Update Admin
Set mobile no='999999999'
Where admin_id= 3;

```
SET mobile_no = '9999999999'
WHERE admin_id = 3;

SELECT * FROM Admin;
```

admin_id	first_name	middle_name	last_name	mobile_no	password
1	Ramesh	NULL	Sharma	9000000001	admin1
2	Suresh	NULL	Patel	9000000002	admin2
3	Anita	NULL	Verma	9999999999	admin3
4	Karan	NULL	Singh	9000000004	admin4
5	Meena	NULL	Joshi	9000000005	admin5
NULL	NULL	NULL	NULL	NULL	NULL

Q11 : change Admin id to 1 for id 2?

Ans : Update Pdf
Set admin -id=1
Where pdf_id= 2;

```
SET admin_id = 1
WHERE pdf_id = 2;

SELECT * FROM PDF;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Cont

pdf_id	title	description	file_path	upload_date	admin_id	category_id
1	DBMS Basics	Intro notes	/files/dbms.pdf	2024-01-10	1	1
2	CN Short Notes	Networking	/files/cn.pdf	2024-01-12	1	2
3	OS Concepts	Processes	/files/os.pdf	2024-01-15	3	3
4	Java Guide	OOP	/files/java.pdf	2024-01-18	4	4
5	Python Handbook	Basics	/files/python.pdf	2024-01-20	5	5
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Q12 : delete pdf id 5?

Ans : Delete from Pdf
Where pdf_id= 5;

```
DELETE FROM PDF
WHERE pdf_id = 5;

SELECT * FROM PDF;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

pdf_id	title	description	file_path	upload_date	admin_id	category_id
1	DBMS Basics	Intro notes	/files/dbms.pdf	2024-01-10	1	1
2	CN Short Notes	Networking	/files/cn.pdf	2024-01-12	1	2
3	OS Concepts	Processes	/files/os.pdf	2024-01-15	3	3
4	Java Guide	OOP	/files/java.pdf	2024-01-18	4	4
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Q13 : delete user id 5?

Ans : Delete from USer
Where user_id= 5;

```
DELETE FROM User
WHERE user_id = 5;

SELECT * FROM User;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

pdf_id	title	description	file_path	upload_date	admin_id	category_id
1	DBMS Basics	Intro notes	/files/dbms.pdf	2024-01-10	1	1
2	CN Short Notes	Networking	/files/cn.pdf	2024-01-12	1	2
3	OS Concepts	Processes	/files/os.pdf	2024-01-15	3	3
4	Java Guide	OOP	/files/java.pdf	2024-01-18	4	4
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Q14 : Display all title from the pdf where id is 1,3,4 ?
Ans : select title from PDF where pdf_id in (1,3,4);

```
SELECT title FROM PDF WHERE pdf_id IN (1,3,4);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

title
DBMS Basics
OS Concepts
Java Guide

Q15 : Display all the pdf where id is not in 2,5?
Ans : select * from PDF where pdf_id not in (2,5);

```
SELECT * FROM PDF WHERE pdf_id NOT IN (2,5);
```

pdf_id	title	description	file_path	upload_date	admin_id	category_id
1	DBMS Basics	Intro notes	/files/dbms.pdf	2024-01-10	1	1
3	OS Concepts	Processes	/files/os.pdf	2024-01-15	3	3
4	Java Guide	OOP	/files/java.pdf	2024-01-18	4	4
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Q16 : Display all the first name from user where name start with r?
Ans : select first_name from User where first_name like ‘R%’;

```
SELECT first_name FROM User WHERE first_name LIKE 'R%';
```

first_name
Rohit
Rahul

Q17 : Display all thetitles in ascending order?
Ans : select title from PDF order by title ;

```
SELECT title FROM PDF ORDER BY title;
```

title
CN Short Notes
DBMS Basics
Java Guide
OS Concepts

Q18 : Display all title from the pdf whereadmin id=2 ?
Ans : select title from PDF where admin_id=2;

```
SELECT title FROM PDF WHERE admin_id = 2;
```

title
CN Short Notes

Q19 : Display all title from the pdf with iaitaion j ?
Ans : select title from PDF title like'j%';

```
SELECT title FROM PDF WHERE title LIKE 'J%';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

title
Java Guide

Q20 : delete login id 5?

Ans : Delete from Access_log
Where login_id= 5;

```
DELETE FROM Access_Log
WHERE login_id = 5;

SELECT * FROM Access_Log;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

login_id	access_date	action	pdf_id	user_id
1	2024-02-01 10:00:00	VIEW	1	1
2	2024-02-01 10:30:00	DOWNLOAD	2	2
3	2024-02-02 11:00:00	VIEW	3	3
4	2024-02-03 12:00:00	DOWNLOAD	4	4
NULL	NULL	NULL	NULL	NULL

Q21 : Display all title from the pdf where upoloded btween 12 to 20 jan 20204 ?

Ans : select title from PDF
Where upload date between ‘2024-01-12’ and ‘2024-01-20’;

```
SELECT title FROM PDF
WHERE upload_date BETWEEN '2024-01-12' AND '2024-01-20';
```

title
CN Short Notes
OS Concepts
Java Guide

Q22 : Display all access log where user id is not null ?

Ans : select * from Access log Where upload user id is not null;

```
SELECT * FROM Access_Log WHERE user_id IS NOT NULL;
```

login_id	access_date	action	pdf_id	user_id
1	2024-02-01 10:00:00	VIEW	1	1
2	2024-02-01 10:30:00	DOWNLOAD	2	2
3	2024-02-02 11:00:00	VIEW	3	3
4	2024-02-03 12:00:00	DOWNLOAD	4	4
NULL	NULL	NULL	NULL	NULL

Q23 : Display all titles from pdf in descending order of upload date ?

Ans : select title from pdf order by upload date desc;

```
SELECT title FROM PDF ORDER BY upload_date DESC;
```

title
Java Guide
OS Concepts
CN Short Notes
DBMS Basics

Q24 : Display all the users ?
Ans : select * from user;

```
SELECT * FROM User;
```

user_id	first_name	middle_name	last_name	mobile_no	password
1	Amit	NULL	Verma	8000000001	newpass
2	Neha	NULL	Singh	8000000002	user2
3	Rohit	NULL	Kumar	8000000003	user3
4	Priya	NULL	Shah	8000000004	user4
5	Rahul	NULL	Mehta	8000000005	user5
NULL	NULL	NULL	NULL	NULL	NULL

Q25 : Display all titles from pdf where admin id=1?
Ans : select title from pdf where admin id=1;

```
SELECT title FROM PDF WHERE admin_id = 1;
```

title
DBMS Basics
CN Short Notes

Conclusion:

R1	R2	R3	R4	Total	Sign with Date
(2)	(4)	(2)	(2)	(10)	